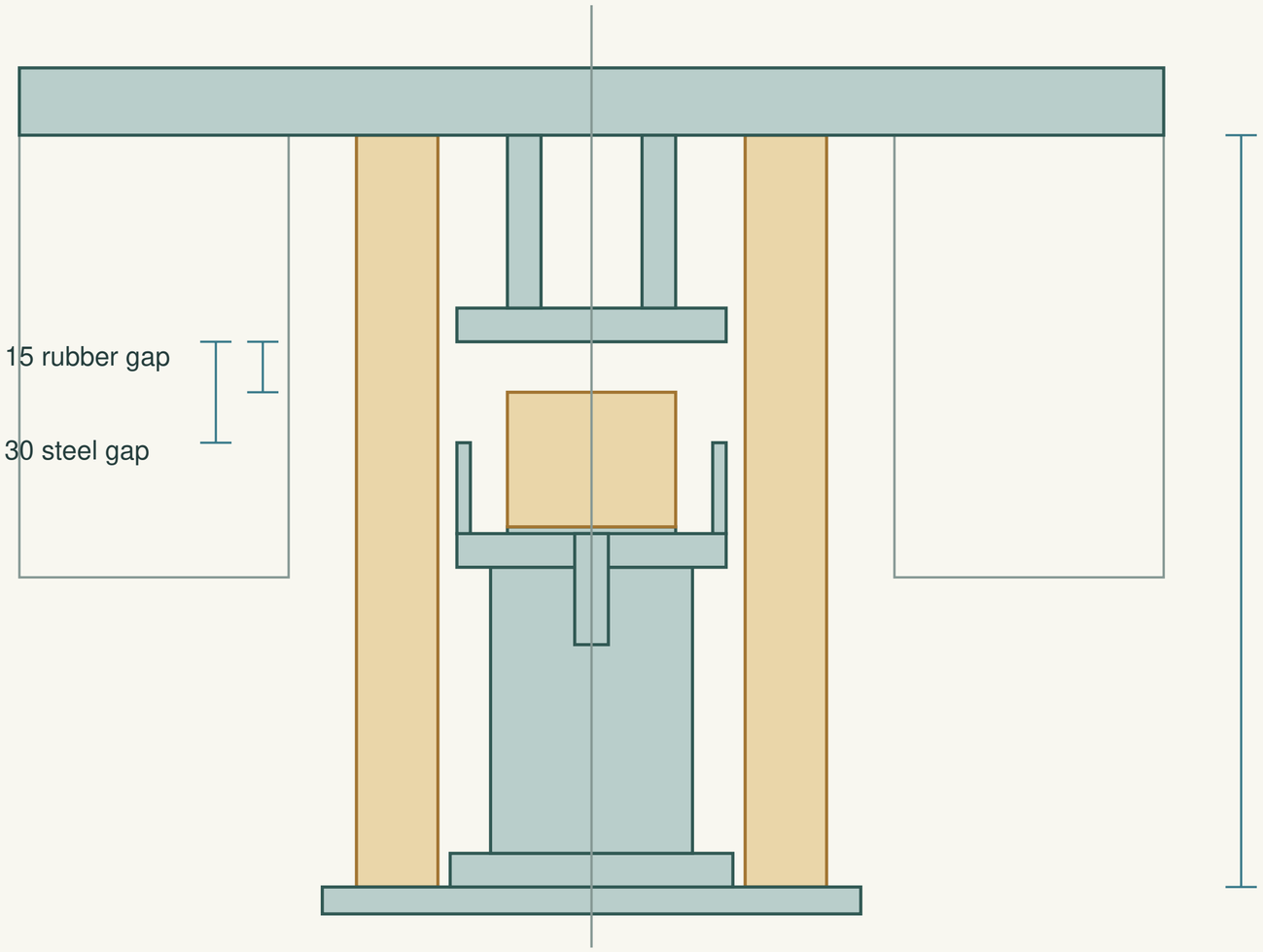
PULSIPHER FARMS · LT34 · COMMERCIAL SPRING STUDY

Eight main springs · D5-Outer

Longitudinal section at axle4 / right side; nominal zero-jounce installation. Same hardware concept at all eight corners.

SECTION X–Z · spring envelopes shown shaded



PURCHASE IDENTIFIERS

8 × D5-Outer rail load spring

COMET100402 / StratoD5-O

Price / eight-unit stock / ETA: quotation

CATALOGUE + CALCULATED BASIS

OD139.700; wire24.209; bore91.281 nominal

Free260.350; solid166.688

Calculated rate392.569 N/mm

Rate =36,768.84 N ÷93.6625 mm

Solid load is not a fatigue working rating.

REVISED LOCAL HARDWARE

Lower seat160 ×160 ×8; topZ858

Cap340 ×160 ×20; undersideZ = table

Enlarge180-wide pocket to cap underside

GMT pedestal baseØ84; rimØ80/72

StrikerØ80 ×10; undersideZ1020

Support tubeØ50/30 to cap underside

GMT STOP-BF1001-71 retained ×8

Same axle preload on each side; actual equilibrium heights differ. Measure each delivered spring before setting.

Shaded gold: supplier packaging envelope. Dashed outline: retained frame beside the enlarged pocket.

Axle-specific heights and cap/rebound attachment plan: LT34-S02. Roll support: LT34-S03.

LT34-S01 · REV A · 23 OCT 2026 · ALL DIMENSIONS mm · DO NOT SCALE

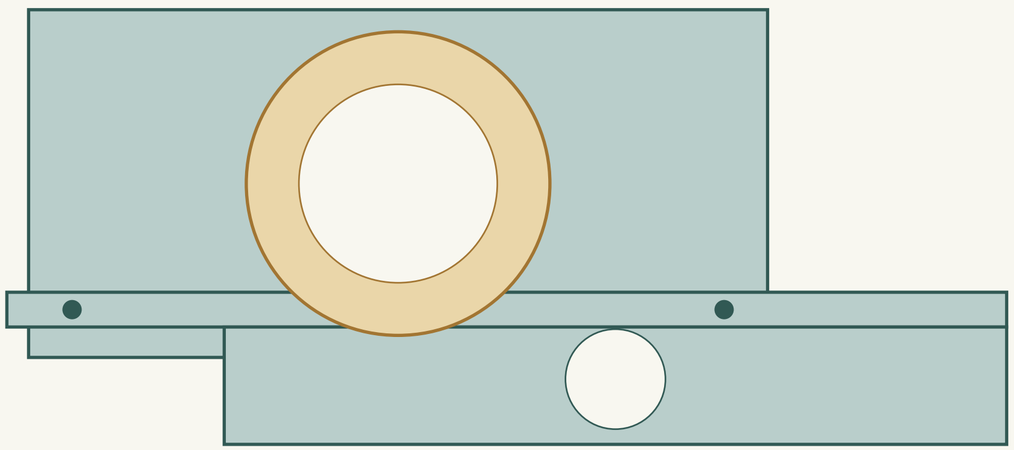
DEVELOPMENT REVIEW ONLY — supplier tolerances, local structure, impact and full assembly clearance remain unreleased.

Main spring seat schedule and rebound attachment

The new cap and rebound bracket form one welded review assembly. Rod, bushing and capture travel positions stay at their prior datums.

Axle / X	Preload N	Datum H	H at +30	H at –20	Cap underside Z	Cap top Z
1 / 3300	12685.6	228.036	198.036	248.036	1086.036	1106.036
2 / 4850	8765.8	238.021	208.021	258.021	1096.021	1116.021
3 / 6500	14309.6	223.899	193.899	243.899	1081.899	1101.899
4 / 8050	14518.2	223.368	193.368	243.368	1081.368	1101.368

Plan · right side · dimensions relative to spring centre (X0 / Y0); mirror for left side



CAP / BRACKET SCHEDULE

Cap:340 ×160 ×20 at axle-specific Z

Rebound web:460 ×16 ×94, X–180..+280

Web Y+50..+66; Z1016..1110

Cartridge shelf:360 ×54 ×16, Z1016..1032

Bushing centre:X+100 / Y+90 (retained)

4 × M16 mounts:X±150; Z1060 and1090

Web clearance:Ø150 around spring below cap

Drill through assembled cap/web at mounts

Weld, thread engagement and fatigue: HOLD

Minimum main spring height at the nominal steel limit:193.368; solid-height margin26.680.

A cap replaces the old70mm pocket header. Gross section equivalence has NOT been demonstrated.

Revised attachment avoids the known old-bracket/cap overlap; full bolt-head, nut, rod and assembly sweep remains to be checked.

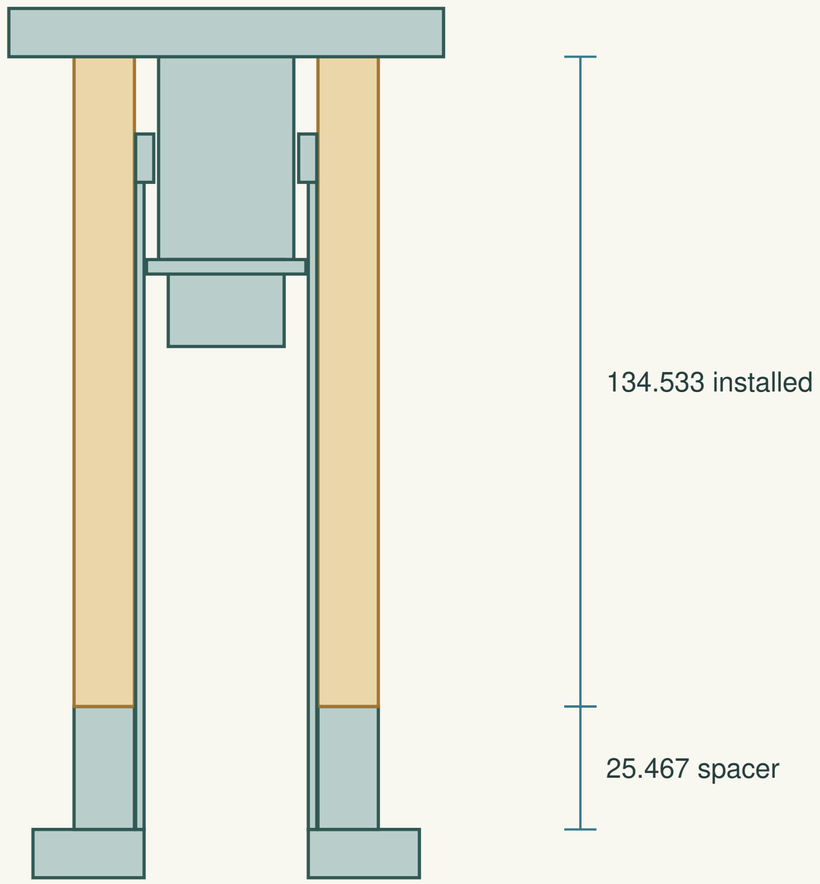
Pocket and hardware CAD is a dimensional review model, not a machining or welding release.

DEVELOPMENT REVIEW ONLY — supplier tolerances, local structure, impact and full assembly clearance remain unreleased.

PULSIPHER FARMS · LT34 · COMMERCIAL SPRING STUDY

Four roll-support springs · Special Springs G63-152

Preferred stocked option; softer458 N/mm rate evaluated with the D5 primary springs. Section through captured cartridge.



PURCHASE BASIS

4 × Special Springs G63-152

Berger:£37.19 each exVAT; £148.76 for4

Listed dispatch27Oct2026, up to5 pieces

Freight / import / arrival date unquoted

SPRING CHECK

Free152; rate458 N/mm ±10%

8,000 N preload ⇒17.467 compression

At +16 capture:118.533 spring height

Total compression33.467 <38 max working

At –16 capture:150.533 spring height

Nominal residual force672 N (still seated)

Long-life reference deflection25.8 mm

SEATS / GUIDE / CAPTURE

SpacerØ63/38 ×25.467 nominal; trim to load

GuideOD37.5 / ID34; bodyZ960..1094

Top lipID30; Z1094..1104 (retained)

Guide wall1.75: compression/retention HOLD

OriginalØ28 plunger andØ33 keeper retained

Spring rod38 ⇒ ≥0.25 radial guide clearance

4M8 onPCD64 and ±16 capture retained

Alternative Bordignon5S63152:801 N/mm; use17.988mm spacer, not this G63-152 spacer.

Set preload with measured free height and load; free-length tolerance±1% is not absorbed by one fixed spacer.

G63-152 at –10% rate, reset8kN preload:35.408mm total at +16, below38mm working limit.

LT34-S03 · REV A · 23 OCT 2026 · ALL DIMENSIONS mm · DO NOT SCALE

DEVELOPMENT REVIEW ONLY — supplier tolerances, local structure, impact and full assembly clearance remain unreleased.